



Rossmoyne Senior High School

Semester One Examination, 2023

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

MARKING GUIDE

Section Two: Calculator-assumed

WA student number: In figures

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In words

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Your name

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Time allowed for this section

Reading time before commencing work: ten minutes

Working time: one hundred minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators, which can include scientific, graphic and Computer Algebra System (CAS) calculators, are permitted in this ATAR course examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	53	35
Section Two: Calculator-assumed	12	12	100	101	65
Total					100

Instructions to candidates

- The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
- You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
- Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
- It is recommended that you do not use pencil, except in diagrams.
- Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
- The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
8	7	
9	7	
10	8	
11	9	
12	7	
13	9	
14	7	
15	10	
16	9	
17	9	
18	8	
19	11	
S2 Total	101	
S2 Wt ($\times 0.6633$)	65%	

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Section Two: Calculator-assumed

65% (101 Marks)

This section has **twelve** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 8

(7 marks)

- (a) Given that $f(x) = 4x^2 + 3x + 1$, determine $f\left(-\frac{1}{3}\right)$. (1 mark)

Solution
$f\left(-\frac{1}{3}\right) = \frac{4}{9}$
Specific behaviours
✓ correct value

- (b) State the natural domain and range for the following functions.

- (i) $f(x) = \frac{3}{x+4}$. (2 marks)

Solution
The domain is $x \neq -4$ and the range is $y \neq 0$.
Specific behaviours
✓ correct domain ✓ correct range

- (ii) $g(x) = 5 - \sqrt{x-2}$. (2 marks)

Solution
The domain is $x \geq 2$ and the range is $y \leq 5$.
Specific behaviours
✓ correct domain ✓ correct range

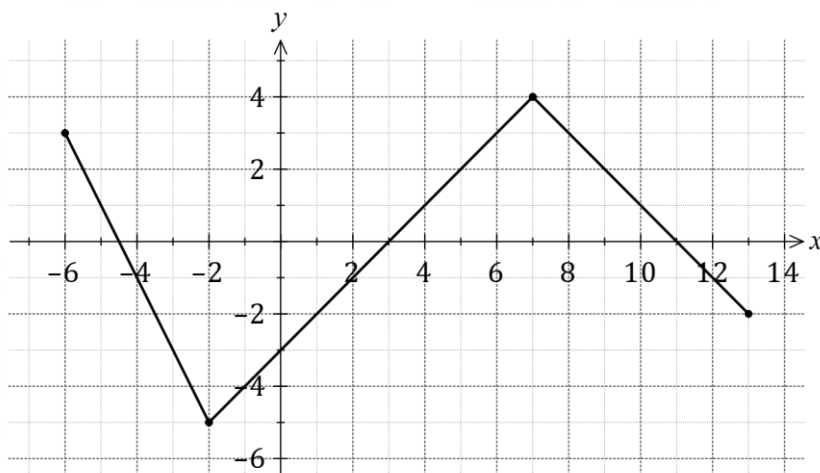
- (c) Determine the range of the function $f(x) = (x-1)^2 - 3$ for the domain $0 \leq x \leq 4$. (2 marks)

Solution
f has a minimum at $(1, -3)$, $f(0) = -2$ and $f(4) = 6$. Hence range is $-3 \leq y \leq 6$.
Specific behaviours
✓ once correct bound of range ✓ correct range

Question 9

(7 marks)

The diagram below shows the graph of $y = f(x)$, where f is a function with domain $-6 \leq x \leq 13$.



Determine the **coordinates** of the following point(s):

- (a) the y -axis intercept of the graph of $y = f(x - 4)$.

(1 mark)

Solution
Graph is translated 4 units right: $(0, -1)$.
Specific behaviours
✓ correct coordinates

- (b) the roots of the graph of $y = f(x) + 1$.

(2 marks)

Solution
Graph is translated 1 units upwards: $(-4, 0)$, $(2, 0)$ and $(12, 0)$.
Specific behaviours
✓ any correct root
✓ three correct roots

- (c) the maximum of the graph of $y = -4f(x)$.

(2 marks)

Solution
Vertical dilation factor -4 , min of $f(x)$ transformed to max of $-4f(x)$: $(-2, 20)$.
Specific behaviours
✓ vertical dilation of any of the four points by factor -4
✓ correct coordinates of minimum

- (d) the minimum of the graph of $y = f\left(\frac{x}{2}\right) - 7$.

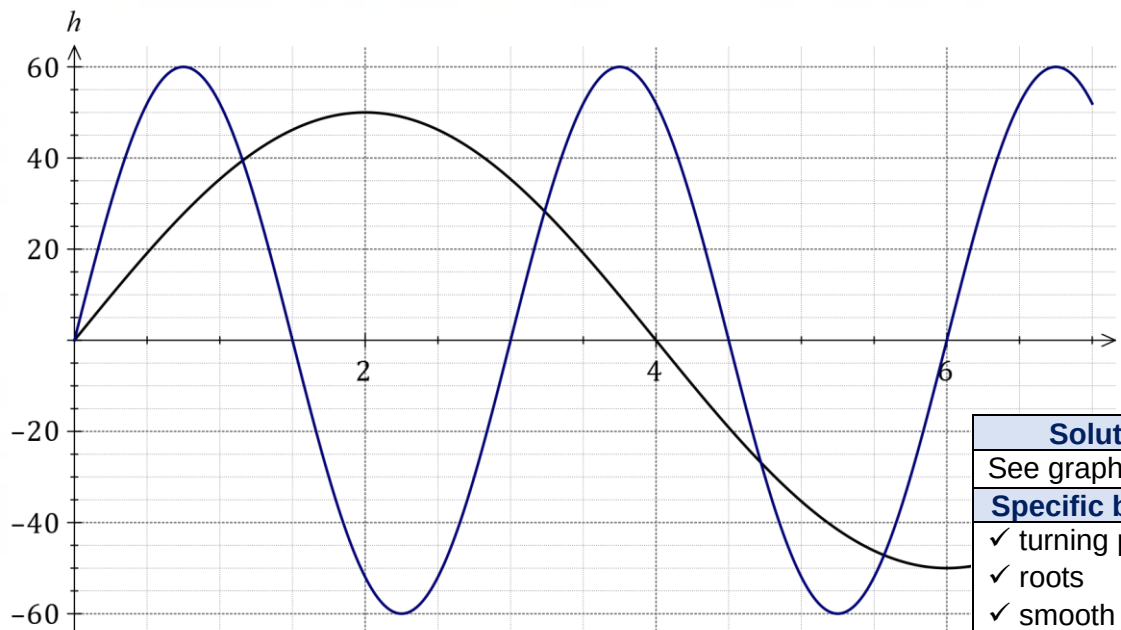
(2 marks)

Solution
Horizontal dilation factor 2 and translation 7 down: $(-4, -12)$.
Specific behaviours
✓ correct x -coordinate
✓ correct y -coordinate

Question 10

(8 marks)

The vertical height h , in metres, of the tip of a wind turbine blade above its centre of rotation at time t seconds, $t \geq 0$, is shown in the graph below.



Solution (c)
See graph
Specific behaviours
✓ turning points
✓ roots
✓ smooth curve

- (a) State the amplitude and period of the trigonometric function h . (2 marks)

Solution
Amplitude: 50 m. Period: 8 seconds.
Specific behaviours
✓ amplitude
✓ period

A similar function for a second wind turbine blade is given by $h_2 = 60 \sin\left(\frac{2\pi t}{3}\right)$, $t \geq 0$.

- (b) State the period of this second function and hence determine how many revolutions per minute this blade is rotating at. (2 marks)

Solution
Period: $\omega = 2\pi \div \frac{2\pi}{3} = 3$ seconds. Rotating at $60 \div 3 = 20$ rpm.
Specific behaviours
✓ period
✓ rotational speed

- (c) Add the graph of function h_2 to the axes above. (3 marks)

- (d) At what time does the tip of the second blade first reach its maximum height? (1 mark)

Solution
First maximum of graph at $3 \div 4 = 0.75$ seconds.
Specific behaviours
✓ correct time

Question 11

(9 marks)

A dodecahedral dice has 12 sides, each numbered with a different integer between 1 and 12 inclusive. When the dice is rolled once, event A occurs if the number is even, event B occurs if the number is less than 8 and event C occurs if the number is a multiple of 3.

- (a) List the sets A , B , and C when the dice is thrown once. (1 mark)

Solution
$A = \{2, 4, 6, 8, 10, 12\}$ $B = \{1, 2, 3, 4, 5, 6, 7\}$ $C = \{3, 6, 9, 12\}$
Specific behaviours
✓ correct description

- (b) Write a brief word description of the numbers in the outcome set $B \cap \bar{A}$. (1 mark)

Solution
Odd numbers less than 8
Specific behaviours
✓ correct description

- (c) List the following sets of outcomes:

(i) $B \cup C$

Solution
$B \cup C = \{1, 2, 3, 4, 5, 6, 7, 9, 12\}$
Specific behaviours
✓ correct elements ✓ use of curly brackets (penalise once in question)

(2 marks)

(ii) $A \cap \bar{B}$.

Solution
$A \cap \bar{B} = \{8, 10, 12\}$
Specific behaviours
✓ correct set

(1 mark)

(iii) $\overline{A \cup B \cup C}$.

Solution
$\overline{A \cup B \cup C} = \{11\}$
Specific behaviours
✓ correct set

(1 mark)

(iv) $A \cap B \cap C$.

Solution
$A \cap B \cap C = \{6\}$

(1 mark)

- (d) Determine $P(\overline{A \cap B} \cap C)$.

Specific behaviours
✓ correct set

(2 marks)

Solution
$\overline{A \cap B} \cap C = \{3, 9, 12\}, \quad P(\overline{A \cap B} \cap C) = \frac{3}{12} = \frac{1}{4}$
Specific behaviours
✓ indicates correct set of outcomes ✓ correct probability

See next page

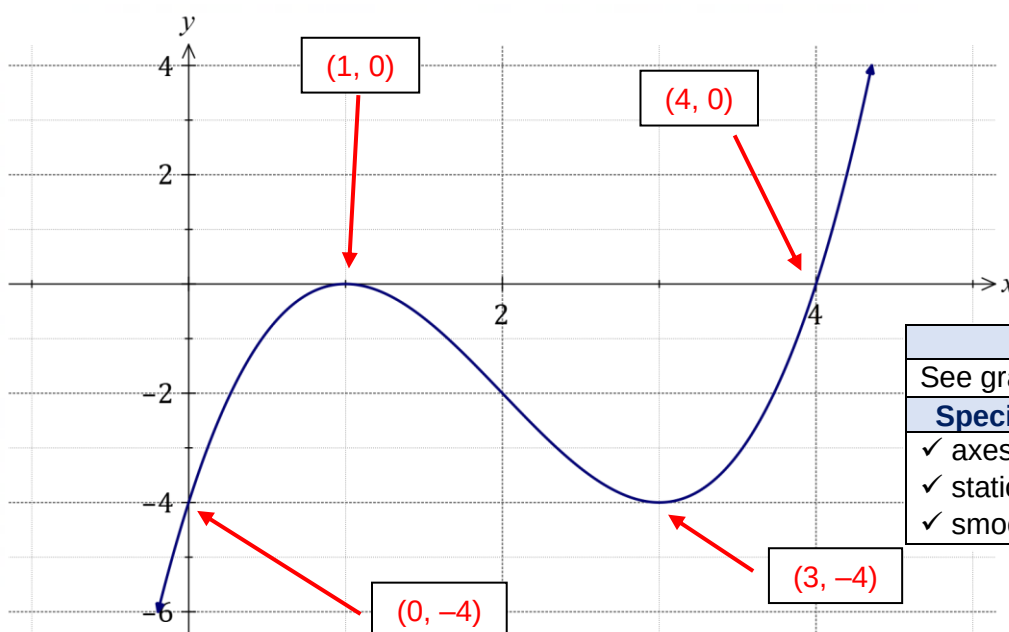
Question 12

(7 marks)

Let $f(x) = x^3 - 6x^2 + 9x - 4$.

(a) Sketch the graph of $y = f(x)$ on the axes below.

(3 marks)



Solution
See graph
Specific behaviours
✓ axes intercepts
✓ stationary points
✓ smooth curve

Let L_1 be the straight line that passes through the local minimum and local maximum of the graph of $y = f(x)$.

(b) Determine the equation of L_1 .

(2 marks)

Solution
Gradient through $(3, -4)$ and $(1, 0)$: $m = \frac{-4}{3-1} = -2$ $y = -2(x - 1)$ $y = -2x + 2$
Specific behaviours
✓ gradient ✓ correct equation

Let L_2 be the straight line through $(0, -22)$ and parallel to L_1 .

(c) Determine x -coordinate of the point of intersection of the graph of $y = f(x)$ and L_2 .

(2 marks)

Solution
Equation of L_2 is $y = -2x - 22$. $x^3 - 6x^2 + 9x - 4 = -2x - 22 \Rightarrow x = -1$
Specific behaviours
✓ forms required equation ✓ correct solution

Question 13

(9 marks)

The graph of the hyperbola $y = \frac{p}{x+q}$, where p and q are constants, passes through the point $(7, 1)$ and intercepts the vertical axis at $y = -4/3$.

- (a) Show how to use this information to deduce that $p = 4$ and $q = -3$. (2 marks)

Solution
Using $(7, 1)$ and $(0, -4/3)$:
$1 = \frac{p}{7+q} \Rightarrow p = q + 7, \quad -\frac{4}{3} = \frac{p}{q} \Rightarrow p = -\frac{4q}{3}$
Hence
$-\frac{4q}{3} = q + 7 \Rightarrow \frac{7q}{3} = -7 \Rightarrow q = -3, p = -3 + 7 = 4.$
Specific behaviours
✓ uses points to form two equations in p and q ✓ indicates appropriate solution method (CAS, algebraic, etc)

- (b) State the equations of the asymptotes of the hyperbola. (2 marks)

Solution
Using CAS to graph $y = \frac{4}{x-3}$ or by inspection, the equation of the horizontal asymptote is $y = 0$ and the equation of the vertical asymptote is $x = 3$.
Specific behaviours
✓ equation of horizontal asymptote ✓ equation of vertical asymptote

The line with equation $y = 16x - 48$ intersects the hyperbola at two points.

- (c) Show how to use the equations of the line and the hyperbola to formulate a quadratic equation in the form $ax^2 + bx + c = 0$, the solutions of which are the x -coordinates of the points of intersection. Do **not** determine these x -coordinates. (3 marks)

Solution
$16x - 48 = \frac{4}{x-3}$
$4x - 12 = \frac{1}{x-3}$
$(4x - 12)(x - 3) = 1$
$4x^2 - 24x + 36 = 1$
$4x^2 - 24x + 35 = 0$
Specific behaviours
✓ forms equations ✓ shows product of binomial terms ✓ expands and simplifies into required form

- (d) Determine the coordinates of the points of intersection of the line and hyperbola. (2 marks)

Solution
Using CAS, they intersect at $(2.5, -8)$ and $(3.5, 8)$.
Specific behaviours
✓ one correct point ✓ second correct point

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Question 14

(7 marks)

The table below shows the results of a random survey of 200 customers who recently bought a new TV from a large store. The customers were asked which brand of TV they bought and whether they would buy the same brand again.

Buy same brand again	TV Brand			
	Sense	Sunny	GL	Other
Yes	14	31	19	24
No	11	15	8	30
Unsure	7	6	10	25

A customer who recently bought a TV from the store is selected at random. Use the above data to determine a point estimate for the probability that

- (a) the TV they bought was a Sunny and that they would not buy the same brand again.

(1 mark)

Solution
$P = \frac{15}{200} \left(= \frac{3}{40} = 0.075 \right)$
Specific behaviours
✓ correct probability

- (b) they are unsure whether they would buy the same brand of TV again.

(1 mark)

Solution
$P = \frac{7 + 6 + 10 + 25}{200} = \frac{48}{200} \left(= \frac{6}{25} = 0.24 \right)$
Specific behaviours
✓ correct probability

- (c) the brand of TV they bought was not GL.

(1 mark)

Solution
$P = \frac{200 - (19 + 8 + 10)}{200} = \frac{200 - 37}{200} = \frac{163}{200} (= 0.815)$
Specific behaviours
✓ correct probability

- (d) they would not buy the same brand of TV again, given that the brand they bought was not 'Other'.

(2 marks)

Solution
$P = \frac{11 + 15 + 8}{200 - (24 + 30 + 25)} = \frac{34}{200 - 79} = \frac{34}{121} (\approx 0.281)$
Specific behaviours
✓ correct denominator
✓ correct probability

- (e) the brand of TV they bought was Sense or GL, given that they would buy the same brand again. (2 marks)

Solution	
$\frac{14 + 19}{14 + 31 + 19 + 24} = \frac{33}{88} \left(= \frac{3}{8} = 0.375 \right)$	
Specific behaviours	
✓ correct denominator before simplification ✓ correct probability	

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Question 15

(10 marks)

- (a) Determine the area of triangle OPQ when $\angle POQ = 134^\circ$, $OP = 19$ cm and $OQ = 24$ cm.

(2 marks)

Solution
$A = \frac{1}{2}(19)(24) \sin 134^\circ$ $= 164 \text{ cm}^2$
Specific behaviours
✓ correct use of area formula ✓ correct area

- (b) Triangle OXY has an area of 175 cm^2 and has sides $OX = 43$ cm and $OY = 25$ cm. Determine the size of angle $\angle XOY$, giving your answer(s) in degrees correct to one decimal place.

(8 marks)

Solution
$175 = \frac{1}{2}(43)(25) \sin \angle XOY$ $\angle XOY = 19^\circ \text{ or } 180^\circ - 19^\circ = 161^\circ$ $XY^2 = 43^2 + 25^2 - 2(43)(25) \cos \angle XOY$ When $\angle XOY = 19^\circ$, $XY = 21.0$ cm and when $\angle XOY = 161^\circ$, $XY = 67.13$ cm. $\frac{\sin \angle OXY}{25} = \frac{\sin 19^\circ}{21.0} \Rightarrow \angle OXY = 22.8^\circ$ $\frac{\sin \angle OXY}{25} = \frac{\sin 161^\circ}{67.13} \Rightarrow \angle OXY = 7.0^\circ$ Hence $\angle OXY = 22.8^\circ$ or 7.0°.
Specific behaviours
✓ correct equation for $\angle XOY$ ✓✓ two correct values for $\angle XOY$ ✓ correct equation for XY ✓ at least one correct value for XY ✓ correct equation for $\angle OXY$ ✓✓ two correct values for $\angle OXY$

Question 16

(9 marks)

Records kept by a café on an industrial estate show that 74% of its customers work in transport and the remainder in other jobs. 45% of the customers who work in transport order chips with their meal, whilst for other customers the figure is 30%.

(a) Determine the probability that a customer who arrives at random at the café

(i) does not work in transport and orders chips with their meal.

(2 marks)

Solution
$P(\bar{T} \cap C) = (1 - 0.74) \times 0.3 = 0.26 \times 0.3 = 0.078 \quad \left(= \frac{39}{500} \right)$
Specific behaviours
✓ shows correct use of probability rule ✓ correct probability

(ii) does not order chips with their meal.

(3 marks)

Solution
$ \begin{aligned} P(\bar{C}) &= P(T \cap \bar{C}) + P(\bar{T} \cap \bar{C}) = 0.74 \times (1 - 0.45) + 0.26 \times (1 - 0.3) \\ &= 0.74 \times 0.55 + 0.26 \times 0.7 \\ &= 0.407 + 0.182 \\ &= 0.589 \end{aligned} $
Specific behaviours
✓ indicates correct probability for $T \cap \bar{C}$ ✓ indicates correct probability for $\bar{T} \cap \bar{C}$ ✓ correct probability

(iii) works in transport given that they order chips with their meal.

(2 marks)

Solution
$ \begin{aligned} P(T C) &= \frac{P(T \cap C)}{P(C)} \\ &= \frac{0.74 \times 0.45}{1 - 0.589} = \frac{0.333}{0.411} = \frac{111}{137} \approx 0.8102 \end{aligned} $
Specific behaviours
✓ shows correct use of conditional probability rule ✓ correct probability

(b) Explain whether the records kept by the café indicate that working in transport and ordering chips with a meal are independent events for their customers.

(2 marks)

Solution
Events are not independent since $P(T) = 0.74$ and $P(T C) = 0.8102$ and if independent, these probabilities should be the same, or at least very close to each other when using relative frequencies to estimate probabilities.
Specific behaviours
✓ state not independent ✓ explanation using difference of any appropriate probabilities

Question 17

(9 marks)

A newsagent has a stock of 36 greeting cards, all different to each other, of which 10 are birthday, 11 are anniversary and the remainder are blank cards.

- (a) Determine the number of different selections of three blank cards that could be made from the newsagent. (2 marks)

Solution
There are $36 - 10 - 11 = 15$ blank cards and $\binom{15}{3} = 455$ selections.
Specific behaviours
✓ reasonable attempt to use combination formula ✓ correct number

- (b) Determine the number of different selections of six cards that could be made from the birthday and anniversary cards stocked by the newsagent. (2 marks)

Solution
There are $10 + 11 = 21$ cards to choose from and $\binom{21}{6} = 54\,264$ selections.
Specific behaviours
✓ reasonable attempt to use combination formula ✓ correct number

- (c) The newsagent just sold two of the 36 greeting cards.

- (i) Determine the number of different selections of two cards that could be sold. (1 mark)

Solution
$\binom{36}{2} = 630$
Specific behaviours
✓ correct number

- (ii) Determine the probability that both were birthday cards. (2 marks)

Solution
There are $\binom{10}{2} = 45$ selections of birthday cards, so $p = \frac{45}{630} = \frac{1}{14} \approx 0.0714$
Specific behaviours
✓ indicates correct approach to problem ✓ correct probability

- (iii) Determine the probability that at least one of them was an anniversary card. (2 marks)

Solution
There are $\binom{25}{2} = 300$ selections with no anniversary card, so $p = 1 - \frac{300}{630} = \frac{330}{630} = \frac{11}{21} \approx 0.5238$
Specific behaviours
✓ indicates correct approach to problem ✓ correct probability

See next page

Question 18

(8 marks)

When a chance experiment is carried out, events X and Y occur with probabilities 0.75 and 0.52 respectively.

- (a) Explain why X and Y cannot be mutually exclusive events. (2 marks)

Solution
For mutually exclusive events, $P(X \cup Y) = P(X) + P(Y) = 0.75 + 0.52 = 1.27$. This is clearly impossible as any probability must lie between 0 and 1.
Specific behaviours
✓ shows $P(X) + P(Y) > 1$ ✓ reasonable explanation

- (b) Determine the following probabilities:

- (i) $P(\bar{X} \cap Y)$ when $P(X \cap \bar{Y}) = 0.44$. (2 marks)

Solution
$P(X \cap Y) = P(X) - P(X \cap \bar{Y}) = 0.75 - 0.44 = 0.31$ $P(\bar{X} \cap Y) = P(Y) - P(X \cap Y) = 0.52 - 0.31 = 0.21$
Specific behaviours
✓ calculates $P(X \cap Y)$ ✓ correct probability

- (ii) $P(X \cup Y)$ when X and Y are independent events. (2 marks)

Solution
For independent events, $P(X \cap Y) = P(X) \times P(Y) = 0.75 \times 0.52 = 0.39$. Hence $P(X \cup Y) = P(X) + P(Y) - P(X \cap Y) = 0.75 + 0.52 - 0.39 = 0.88$.
Specific behaviours
✓ calculates $P(X \cap Y)$ ✓ correct probability

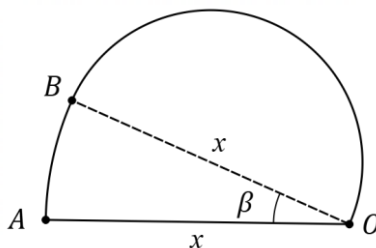
- (iii) $P(\overline{X \cup Y})$ when $P(Y | X) = 0.54$. (2 marks)

Solution
$P(Y X) = \frac{P(X \cap Y)}{P(X)} \Rightarrow P(X \cap Y) = 0.75 \times 0.54 = 0.405$ $P(\overline{X \cup Y}) = 1 - (0.75 + 0.52 - 0.405) = 1 - 0.865 = 0.135$
Specific behaviours
✓ calculates $P(X \cap Y)$ ✓ correct probability

Question 19

(11 marks)

The figure shown in the diagram consists of sector OAB of circle with centre O and a semi-circle with diameter OB .



Let $OA = OB = x$ and $\angle AOB = \beta$ radians.

(a) Let $x = 12$ cm and the area of the semi-circle be **three** times the area of the sector.

(i) Show algebraically that β is equal to $\frac{\pi}{12}$. (3 marks)

Solution
Let area of sector be A_s and of semi-circle be A_c .
$A_s = \frac{1}{2} \times 12^2 \beta = 72\beta, \quad A_c = \frac{1}{2} \pi \left(\frac{12}{2}\right)^2 = 18\pi$
$3A_s = A_c$
$3 \times 72\beta = 18\pi$
$\beta = \frac{18\pi}{3 \times 72} = \frac{\pi}{12}$
Specific behaviours
✓ expressions for area of sector and semi-circle ✓ forms equation using ratio of areas ✓ correct value of β in terms of π

(ii) Determine the perimeter of the figure. (3 marks)

Solution
$P = OA + AB + BO$
$= x + x\beta + \frac{1}{2}\pi x$
$= 12 + 12 \times \frac{\pi}{12} + \frac{1}{2}\pi \times 12$
$= 12 + 7\pi \text{ cm}$
$\approx 33.99 \text{ cm}$
Specific behaviours
✓ Sets up equation ✓ correct arc length AB ✓ correct perimeter

- (b) By first determining a new value for β , determine the area of the figure when the area of the semi-circle is **five** times the area of the sector and the new perimeter of the figure is 77.2 cm. (5 marks)

Solution
$A_s = \frac{1}{2}x^2\beta, \quad A_c = \frac{1}{2}\pi\left(\frac{x}{2}\right)^2 = \frac{x^2\pi}{8}$
$5A_s = A_c$
$5 \times \frac{1}{2}x^2\beta = \frac{x^2\pi}{8}$
$\beta = \frac{\pi}{20}$
$P = x + x\beta + \frac{1}{2}\pi x$
$77.2 = x\left(1 + \frac{\pi}{20} + \frac{\pi}{2}\right)$
$x = 28.3$
$A_c = \frac{28.3^2 \times \pi}{8} = 314.5$
$A_c + A_s = 314.5 + \frac{314.5}{5} = 377 \text{ cm}^2$
Hence area of figure is 377 cm².
Specific behaviours
✓ forms equation for β using ratio of areas ✓ solves for β ✓ forms equation for x using perimeter ✓ solves for x ✓ correct area of figure

Supplementary page

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